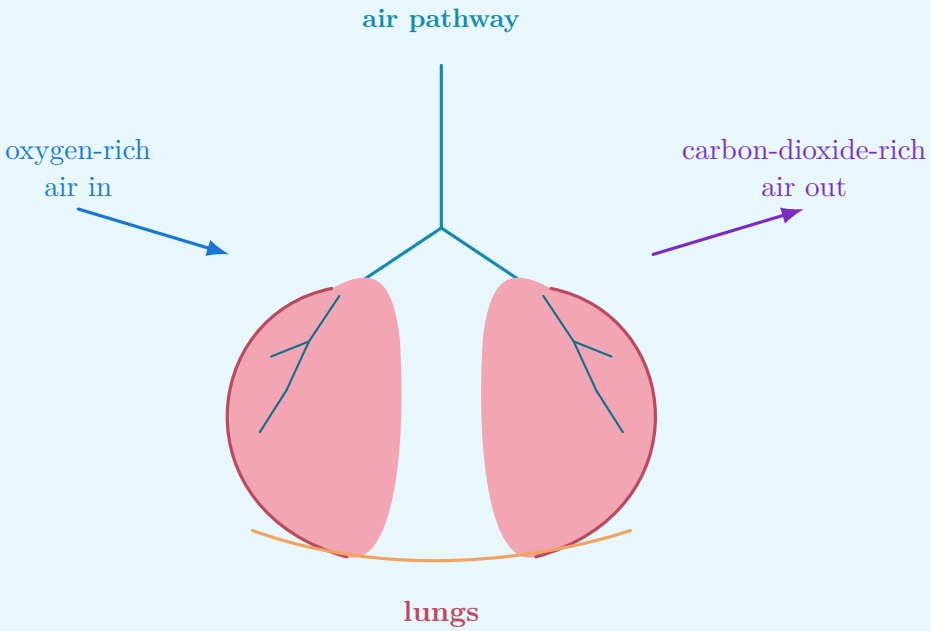
# RESPIRATORY SYSTEM

& Respiration in Animals

Class 7 Science — Life Processes in Animals




**Based on the uploaded Grade 7 *Curiosity* textbook, Chapter 9.**
Main focus: **human respiratory system, breathing, gas exchange and respiration**.
Includes **all 10 end-of-chapter exercise answers first**; digestion-only questions are clearly marked as chapter-completeness items.

# Contents

1 1. Textbook Exercise — Complete Answers 2

2 2. Breathing and Respiration Are Not the Same 3

3 3. Human Respiratory System: Parts and Air Pathway 4

3.1 Simple respiratory-system sketch . . . . . 4

4 4. How Breathing Happens 4

4.1 Inhalation and exhalation . . . . . 5

4.2 Activity 9.2: Bottle-and-balloon model . . . . . 5

5 5. What Do We Breathe Out? 5

5.1 Activity 9.3: Lime-water evidence . . . . . 5

5.2 Inhaled and exhaled air . . . . . 6

6 6. Gas Exchange in the Alveoli 6

7 7. Respiration Releases Energy 6

8 8. Do All Animals Breathe the Same Way? 7

9 9. Health and Everyday Connections 7

10 10. In-Chapter Questions and Activity Answers 7

11 11. Exploratory Projects — Student-Ready Guidance 8

12 12. One-Page Concept Map 8

13 13. Common Mistakes to Avoid 9

14 14. Rapid Revision Test 9

# 1. Textbook Exercise — Complete Answers

**Note:** Questions 1, 2 and 9 are digestion questions. They are included here only so this Chapter 9 exercise set is complete; the main notes remain respiratory-system focused.

### Q1. Complete the journey of food through the alimentary canal

**Food → Mouth → Oesophagus (food pipe) → Stomach → Small intestine → Large intestine → Rectum → Anus.**

### Q2. Chapati / chewed chapati / potato + iodine

- **A: pieces of chapati** — turn **blue-black** because starch is present.
- **B: chewed chapati** — may show **no blue-black colour or only a faint blue-black colour** because saliva breaks down much of the starch into simpler sugars.
- **C: boiled mashed potato** — turns **blue-black** because potato contains starch.

### Q3. Role of the diaphragm in breathing

**Correct option: (iii) To help in inhalation and exhalation.**
During inhalation it moves downward; during exhalation it moves upward.

### Q4. Match the respiratory parts with their functions

| Part | Correct function |
|---|---|
| **Nostrils** | (a) Fresh air from outside enters |
| **Nasal passages** | (d) Tiny hair and mucus trap dust and dirt |
| **Windpipe** | (e) Air reaches our lungs through this part |
| **Alveoli** | (b) Exchange of gases occurs |
| **Rib cage** | (c) Protects lungs |

### Q5. How can Sanvi show that breathing and respiration are different?

She can ask questions such as:

- "Does breathing only move air in and out, while respiration releases energy from glucose?"
- "Is breathing a physical process but respiration a chemical process?"
- "Does respiration use oxygen to break down glucose and produce carbon dioxide, water and energy?"

These questions show that the two processes are connected but **not the same**.

### Q6. What do we inhale?

**Tanu is the most scientifically complete: "We inhale air rich in oxygen."**
Air is a mixture of gases. Inhaled air contains about **21% oxygen**; it is not pure oxygen.

### Q7. Why do we sneeze after inhaling dust-laden air?

Dust particles can irritate the nasal passages. Hair and mucus trap many particles, and sneezing helps force irritating material out of the nose. Thus, sneezing acts as a protective response.

**Q8. Why might Anusha breathe faster than Paridhi after running?**

Possible explanations include:

1. Anusha may have run **faster or with greater effort**, so her body needed more oxygen and produced more carbon dioxide.
2. The two students may differ in **fitness, stamina or recovery rate**; Anusha may need more breaths per minute to meet the body's demand after exercise.

**Core idea:** stronger activity generally increases the body's demand for oxygen and removal of carbon dioxide.

**Q9. Rice flour experiment with saliva**

Yadu wants to test the **action of saliva on starch**.
Test tube A (without saliva) should show a stronger blue-black iodine colour because starch remains.
Test tube B (with saliva and enough time) should show little or no blue-black colour because saliva breaks down starch into simpler sugars.

**Q10. Lime water with inhaled/surrounding air vs exhaled air**

Rakshita is investigating whether **exhaled air contains more carbon dioxide than the surrounding air we inhale**.
She can confirm this by comparing the lime water:

- Test tube A, with surrounding air, shows little/no cloudiness.
- Test tube B, with exhaled air, turns **milky/cloudy more clearly**.

Since carbon dioxide turns lime water milky, this supports her conclusion.

## 2. Breathing and Respiration Are Not the Same

**Key Idea**

**Breathing** is the physical process of **inhaling** air into the lungs and **exhaling** air out of the lungs. **Respiration** is a chemical process inside the body in which oxygen helps break down glucose to release energy.

| **Feature** | Breathing | Respiration |
|---|---|---|
| **Type** | Physical process | Chemical process |
| **Main role** | Brings in oxygen-rich air and removes air with more carbon dioxide | Releases usable energy from glucose |
| **Main place discussed** | Respiratory system, especially lungs | Inside the body/cells |
| **Word equation** | — | Glucose + Oxygen → Carbon dioxide + Water + Energy |

**Memory Trick**

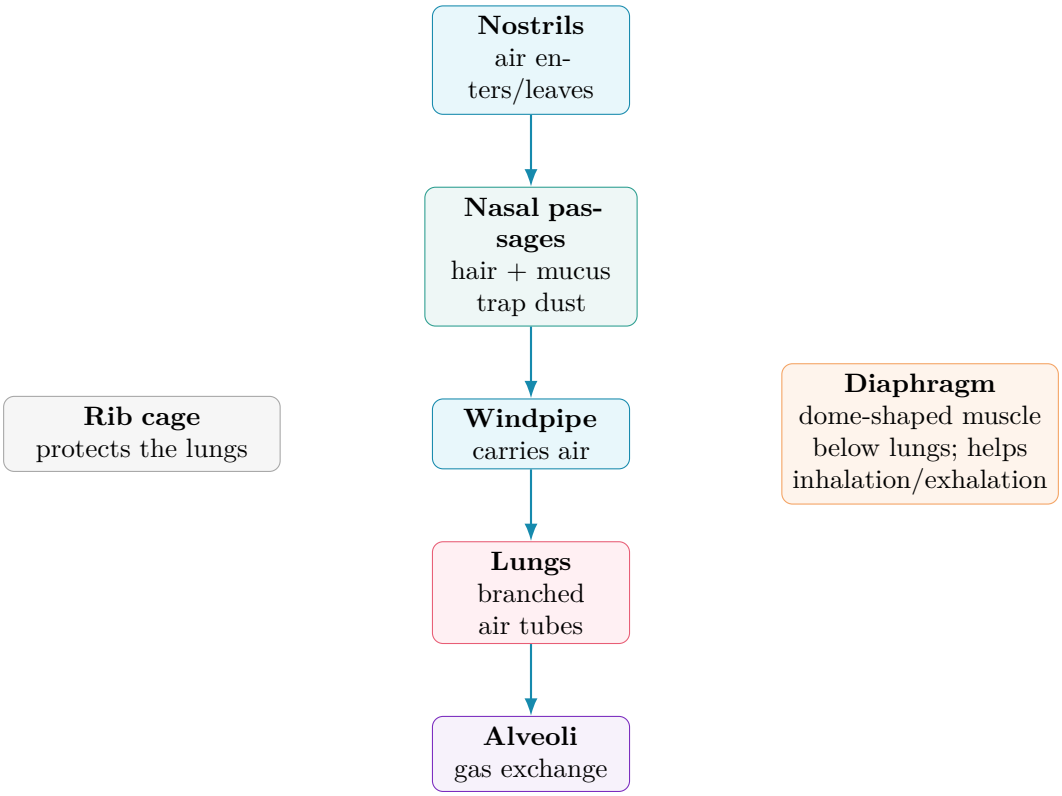
**Breathe = move air. Respire = release energy.** Breathing supplies oxygen; respiration *uses* oxygen.

**Common Mistake**

Do not write "we inhale oxygen". We inhale **air rich in oxygen**. Air also contains other gases.

## 3. Human Respiratory System: Parts and Air Pathway

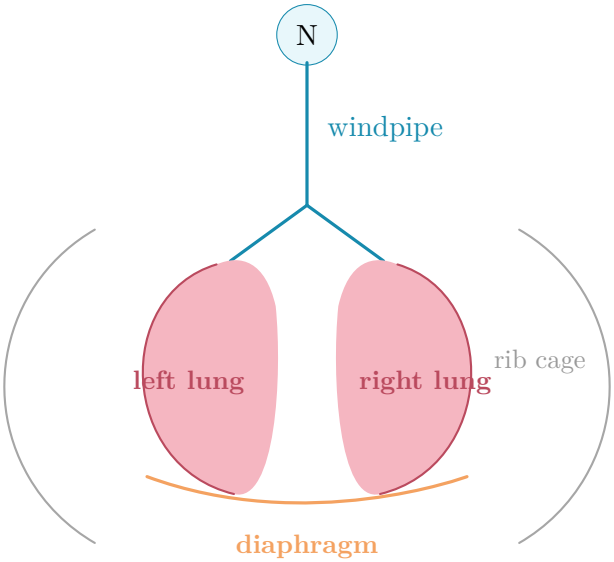
The respiratory system provides a pathway for inhaled and exhaled air and enables exchange of oxygen and carbon dioxide.




**Why is nose-breathing preferred?**

Tiny hair and mucus in the nasal passages help trap dust and dirt before the air reaches the lungs.

### Simple respiratory-system sketch




## 4. How Breathing Happens

### Inhalation and exhalation

| | Inhalation (breathe in) | Exhalation (breathe out) |
|---|---|---|
| **Ribs** | Move up and outwards | Move down and inwards |
| **Diaphragm** | Moves downward | Moves upward |
| **Chest space** | Increases | Decreases |
| **Air** | Enters lungs | Is pushed out |




### Activity 9.2: Bottle-and-balloon model

**Activity / Evidence**

In the textbook model:

- the **two balloons represent the lungs**;
- the **rubber sheet represents the diaphragm**;
- pulling the rubber sheet **down** makes the balloons inflate;
- letting the rubber sheet move **up** makes the balloons deflate.

This models how diaphragm movement changes the space in the chest and helps move air.

**Memory Trick**

**Down = Draw air in. Up = air goes out.**
Diaphragm down → more chest space → inhalation.

## 5. What Do We Breathe Out?

### Activity 9.3: Lime-water evidence

The textbook compares surrounding air with exhaled air using lime water.

**Test tube A**
surrounding air passed through lime water
**little/no visible cloudiness**

**Test tube B**
exhaled air blown through lime water
**turns milky/cloudy**

**Key Idea**

Lime water turns milky in the presence of carbon dioxide. Therefore, the activity shows that **exhaled air contains more carbon dioxide than inhaled/surrounding air**.

### Inhaled and exhaled air

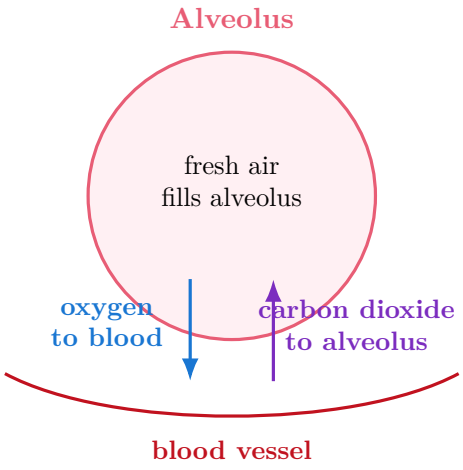
| Gas | Inhaled air | Exhaled air |
|---|---|---|
| **Oxygen** | Nearly 21% | Nearly 16–17% |
| **Carbon dioxide** | Nearly 0.04% | Nearly 4–5% |

**Common Mistake**

Exhaled air is **not oxygen-free**. It still contains about 16–17% oxygen according to the textbook figure.

## 6. Gas Exchange in the Alveoli

**Alveoli** are tiny balloon-like air sacs at the ends of the fine branches inside the lungs. They have thin walls and are surrounded by fine blood vessels.




**Key Idea**

**Oxygen** moves from the alveoli into the blood. **Carbon dioxide** moves from the blood into the alveoli and is later exhaled.

## 7. Respiration Releases Energy

The oxygen carried by blood is used to help break down glucose and release energy.

$$\textbf{Glucose} + \textbf{Oxygen} \longrightarrow \textbf{Carbon dioxide} + \textbf{Water} + \textbf{Energy}$$

This energy helps us walk, run, play and even think. The circulatory system helps transport oxygen and nutrients to body parts and carries waste products away.

**Memory Trick**

**Air pathway** gets oxygen to the lungs. **Blood pathway** carries oxygen from the lungs to the rest of the body.

## 8. Do All Animals Breathe the Same Way?

Different animals use different breathing structures suited to their habitats.

| Animal / group | Main breathing structure described in the textbook |
|---|---|
| **Humans, birds, elephants, lions, cows, goats, lizards, snakes** | Lungs |
| **Most aquatic animals such as fish** | Gills |
| **Tadpoles** | Gills |
| **Adult frogs** | Lungs on land; skin for gas exchange in water |
| **Earthworms** | Moist skin |

**Adaptation idea**

The textbook conclusion is that different animals have different breathing mechanisms adapted to their habitats.

## 9. Health and Everyday Connections

**Health Link**

The textbook notes that smoking damages the lungs and increases the risk of serious respiratory illness. Smoke can also harm non-smokers through passive smoking.

**Breathing practices**

The chapter mentions practices such as pranayama, Tummo breathing and deep breathing in the context of respiratory health, relaxation and concentration.

**Why do we sneeze in dusty air?**

Dust irritates the lining of the nasal passages. Sneezing is a protective response that helps expel irritating particles; nasal hair and mucus also help trap dust.

## 10. In-Chapter Questions and Activity Answers

**What do the balloons and rubber sheet in Fig. 9.9 represent?**

Balloons = **lungs**; rubber sheet = **diaphragm**.

**What happens when the rubber sheet is pulled down?**

The balloons inflate. This models inhalation: diaphragm moves down, chest space increases and air enters the lungs.

**What happens when the rubber sheet moves upward?**

The balloons deflate. This models exhalation: chest space decreases and air is pushed out.

**Why does exhaled air turn lime water milky faster?**

Because exhaled air contains **more carbon dioxide** than inhaled/surrounding air.

**How does gas exchange happen?**

Oxygen passes from the alveoli into the blood, while carbon dioxide passes from the blood into the alveoli to be exhaled.

## 11. Exploratory Projects — Student-Ready Guidance

The textbook gives open-ended projects. The notes below separate what is directly supported by the chapter from useful extension guidance.

**Project: Air quality / AQI and respiratory health**

**Chapter link:** inhaled air reaches the lungs; dust and small particles can affect the respiratory system.
**Student task:** compare how workers such as farmers, factory workers and street vendors may be exposed to dust, smoke or polluted air. Record likely sources of exposure and possible protective practices.
*For actual AQI values or health effects, use a current trusted source; the textbook does not provide AQI ranges.*

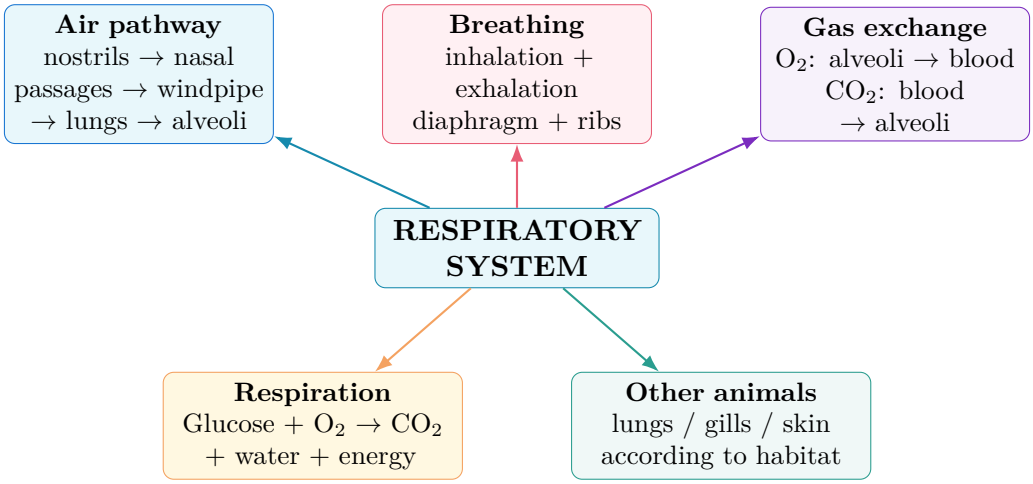
**Project: Box breathing**

**Textbook prompt:** read about box breathing and its benefits.
A simple pattern often used for learning is: breathe in, hold, breathe out, hold for equal counts. This is an **extension beyond the textbook's detailed content**; use a teacher-approved source for benefits and safe practice.

**Project: Why can birds fly at high altitudes?**

The textbook asks students to investigate how bird respiratory systems may be adapted for low-oxygen conditions. This is a **research question**; the chapter itself does not provide the detailed mechanism. Use a trusted biology source and clearly label findings as extension research.

## 12. One-Page Concept Map




**Memory Trick**

**5-step story:** Air enters → reaches alveoli → oxygen enters blood → oxygen helps release energy from glucose → carbon dioxide returns to lungs and is exhaled.

## 13. Common Mistakes to Avoid

**Common Mistake**

- **Wrong:** Breathing and respiration are identical. **Correct:** breathing is physical; respiration is chemical.
- **Wrong:** We inhale pure oxygen. **Correct:** we inhale air rich in oxygen.
- **Wrong:** Exhaled air contains no oxygen. **Correct:** it still has about 16–17% oxygen.
- **Wrong:** Gas exchange happens in the windpipe. **Correct:** gas exchange occurs in the alveoli.
- **Wrong:** Diaphragm moves up during inhalation. **Correct:** it moves down during inhalation.
- **Wrong:** All animals breathe only through lungs. **Correct:** fish use gills; earthworms use moist skin; frogs use different structures at different stages/conditions.

## 14. Rapid Revision Test

1. What is breathing? *Inhaling and exhaling air.*
2. Name the pair of openings through which air enters. *Nostrils.*
3. What traps dust in the nasal passages? *Tiny hair and mucus.*
4. Which tube carries air toward the lungs? *Windpipe.*
5. Where does gas exchange occur? *Alveoli.*
6. What happens to the diaphragm during inhalation? *It moves downward.*
7. Which gas is higher in exhaled air than inhaled air? *Carbon dioxide.*
8. Write the respiration equation. *Glucose + Oxygen → Carbon dioxide + Water + Energy.*
9. Is breathing physical or chemical? *Physical.*
10. Is respiration physical or chemical? *Chemical.*
11. How do fish breathe? *Through gills.*
12. How do earthworms exchange gases? *Through moist skin.*

**Learn the air pathway + diaphragm movement + alveoli gas exchange + respiration equation, and most respiratory-system questions become easy to reason out.**